The Parallel Multiple Mediator Model

Dr. Shonn Cheng

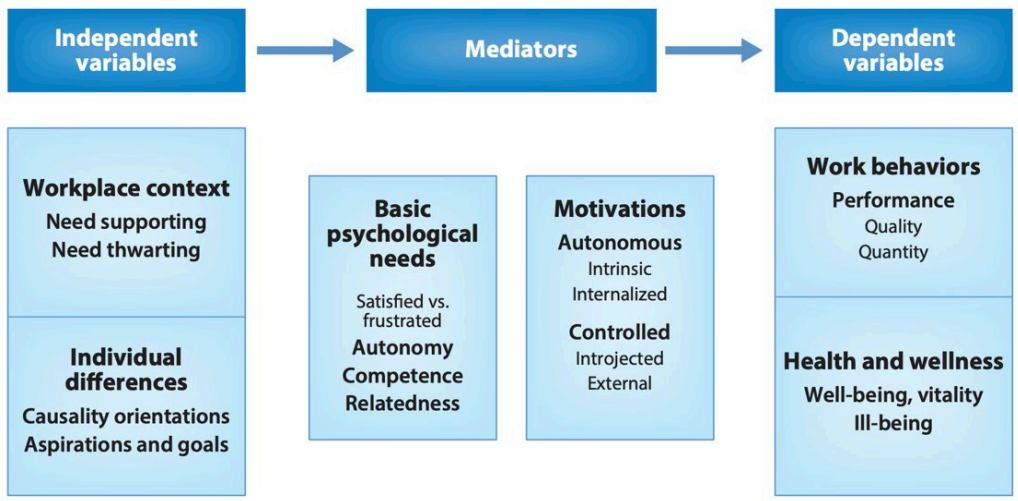
2026-08-01

Problem

Why is it that some people have higher task performance whereas others don't?

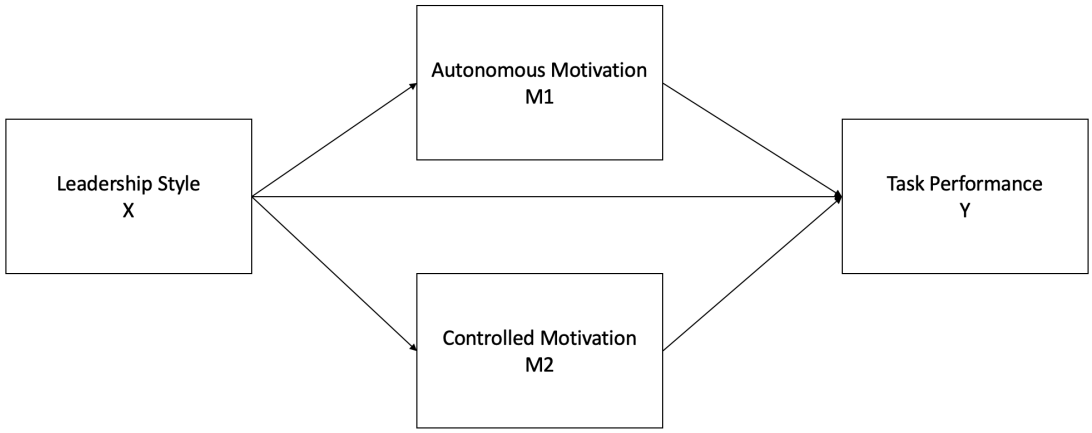
Theory

According to self-determination theory, individuals' work behaviors are predicted by their motivations (autonomous and controlled), which, in turn, are predicted by workplace context (e.g., leadership style).



We want to test if this part of this model is true or not. So, we go out to collect some data but what should we do next?

Notation: Diagram

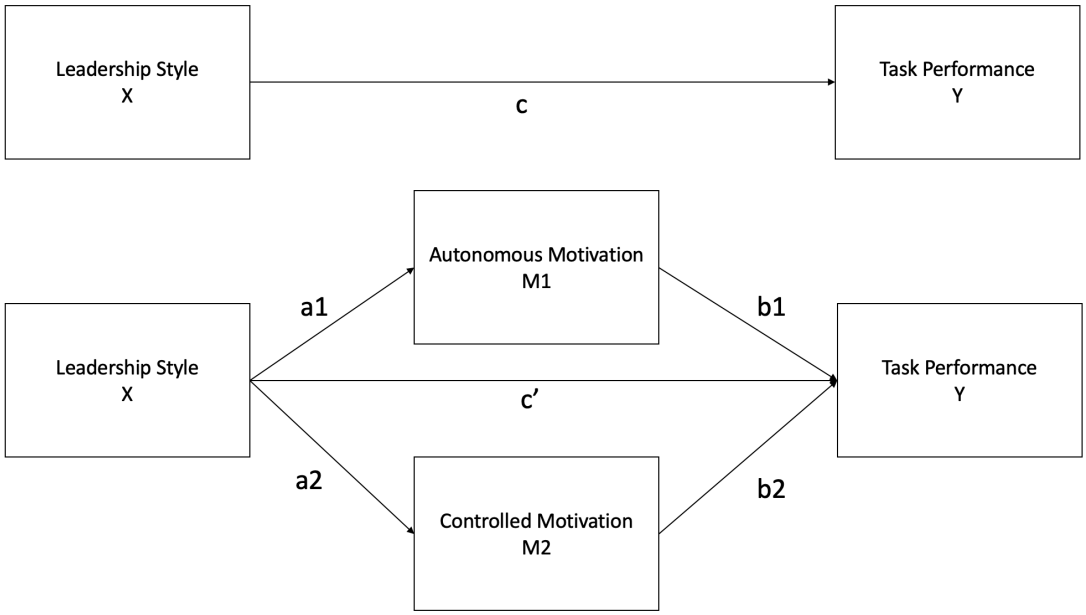


- **X**: causal antecedent variable
- **Y**: consequent variable
- **M1** "mediator 1"
- **M2** "mediator 2"

Activity

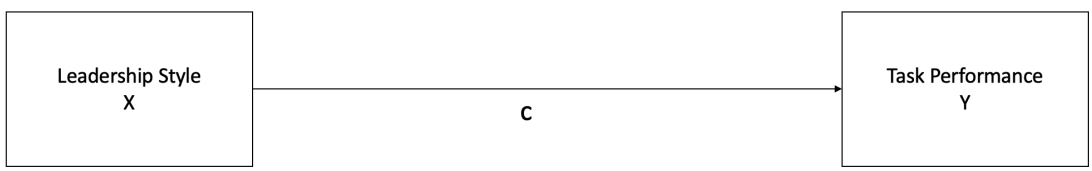
- Use the notation shown here to draw the relation you would like to investigate.

Notation: Total, direct, and indirect effects

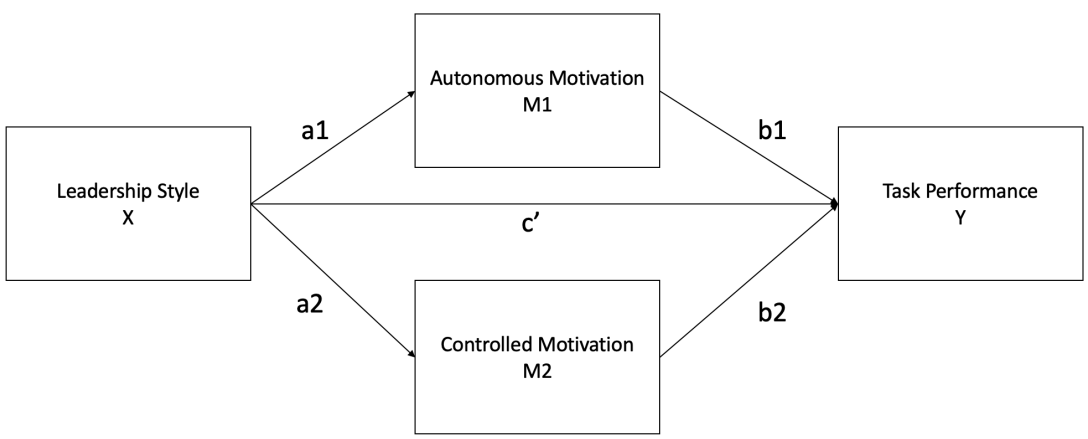


- Let a1, a2, b1, b2, c, and c' be quantifications of causal effects, such as regression coefficients in an ordinary least squares model (or maximum likelihood path estimates in a structural equation model).
- c = "total effect" of X on Y (i.e., the effect of X on Y when we don't account for the effect of M)
- **a1 × b1 = "indirect effect" of X on Y (This is our main focus.)**
- **a2 × b2 = "indirect effect" of X on Y (This is our main focus.)**
- c' = "direct effect" of X on Y
- total effect (c) = direct effect (c') + indirect effect (a1 × b1) + indirect effect (a2 × b2)
- indirect effect (a1 × b1) + indirect effect (a2 × b2) = total effect (c) - direct effect (c')
- We can still do our mediation analysis even if the total effect is zero, **c is not the precursor of the mediation analysis**.

Regression Equation



$$TaskPerformance_j = i + cLeadershipStyle_j + \varepsilon_j$$



$$AutonomousMotivation_j = i + a1LeadershipStyle_j + \varepsilon_j$$

$$ControlledMotivation_j = i + a2LeadershipStyle_j + \varepsilon_j$$

$$TaskPeferomance_j = i + b1AutonomousMotivation_j + b2ControlledMotivation_j + c'LeadershipStyle_j + \varepsilon_j$$

Transformational Leadership

Transformational leadership is a leadership style that can inspire positive changes in those who follow. Transformational leaders are generally energetic, enthusiastic, and passionate. Not only are these leaders concerned and involved in the process, but they are also focused on helping every member of the group succeed.

Run Process Macro

Whenever you would like to run conditional process analysis, don't forget to open and run the script of Process Macro!

```
******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

PROCESS is now ready for use.
Copyright 2020-2023 by Andrew F. Hayes ALL RIGHTS RESERVED
Workshop schedule at http://haskayne.ucalgary.ca/CCRAM
```

Load Packages and Import Data

```r
library(tidyverse)
library(psych)
library(haven)

data <- read_sav("sdt.sav")

#view our data
data
```

```
# A tibble: 500 × 4
   leader  auto control performance
    <dbl> <dbl>   <dbl>       <dbl>
 1   2.15  3.00   2.26         5.29
 2   4.15  4.22   1.70         7.49
 3   2.64  3.16   1.74         4.87
 4   4.53  3.88   1.99         5.91
 5   4.76  4.59   1.84         7.45
 6   1.18  2.07   3.25         3.09
 7   3.11  3.94   2.68         5.60
 8   4.57  4.75   0.849        7.20
 9   3.21  4.81   1.21         6.30
10   2.83  3.38   2.19         5.26
# ℹ 490 more rows
```

```r
#descriptive statistics
describe(data)
```

```
            vars   n mean   sd median trimmed  mad  min  max range  skew
leader         1 500 2.98 1.14   2.91    2.97 1.43 1.00 5.00  4.00  0.10
auto           2 500 3.79 0.91   3.81    3.78 1.02 1.62 5.86  4.25  0.03
control        3 500 2.20 0.89   2.22    2.21 0.92 0.09 4.40  4.31 -0.07
performance    4 500 5.41 1.47   5.34    5.37 1.70 1.89 9.10  7.20  0.17
            kurtosis   se
leader         -1.19 0.05
auto           -0.66 0.04
control        -0.54 0.04
performance    -0.72 0.07
```

Visualize (Bivariate Correlations)

```r
data |> as.matrix() |> Hmisc::rcorr()
```

```
            leader  auto control performance
leader        1.00  0.77   -0.77        0.89
auto          0.77  1.00   -0.60        0.81
control      -0.77 -0.60    1.00       -0.76
performance   0.89  0.81   -0.76        1.00

n= 500



P
            leader auto control performance
leader              0    0       0
auto         0           0       0
control      0      0            0
performance  0      0    0
```

```r
#data |> select(leader, auto, control, performance) |> as.matrix() |> Hmisc::
rcorr()
```

Model (conditional process analysis: the parallel multiple mediator model)

Unfortunately, there is no existing model we can choose to run the parallel multiple mediator analysis. However, the conditional process analysis framework allows us to create and customize our own model. This is how:

	X	M1	M2
M1	1		
M2	1	0	
Y	1	1	1

```
process(data=data, y="performance", x="leader", m=c("auto", "control"), total
=1, seed=12345, bmatrix=c(1, 1, 0, 1, 1, 1))


******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

Model : CUSTOM
    Y : performance
    X : leader
   M1 : auto
   M2 : control

Sample size: 500

Custom seed: 12345


**************************************************************************
Outcome Variable: auto

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.7717     0.5955     0.3384   733.1251     1.0000   498.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.9396     0.0730    26.5599     0.0000     1.7961     2.0831
leader       0.6198     0.0229    27.0763     0.0000     0.5748     0.6647

**************************************************************************
Outcome Variable: control

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.7708     0.5942     0.3216   729.0724     1.0000   498.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     3.9953     0.0712    56.1201     0.0000     3.8554     4.1351
leader      -0.6025     0.0223   -27.0013     0.0000    -0.6463    -0.5587

**************************************************************************
Outcome Variable: performance
```

```
Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.9213     0.8488     0.3303   928.2126     3.0000   496.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     2.1744     0.2151    10.1075     0.0000     1.7518     2.5971
leader       0.6782     0.0445    15.2511     0.0000     0.5909     0.7656
auto         0.4890     0.0443    11.0413     0.0000     0.4020     0.5760
control     -0.2918     0.0454    -6.4238     0.0000    -0.3811    -0.2026

************************ TOTAL EFFECT MODEL **************************
Outcome Variable: performance

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.8933     0.7980     0.4396 1967.3110     1.0000   498.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.9569     0.0832    23.5114     0.0000     1.7934     2.1205
leader       1.1571     0.0261    44.3544     0.0000     1.1059     1.2084

**********************************************************************
Bootstrapping progress:

  |
  |                                                                      |   0%
  |
  |                                                                      |   1%
  |
  |>                                                                     |   1%
  |
  |>                                                                     |   2%
  |
  |>>                                                                    |   2%
  |
  |>>                                                                    |   3%
  |
  |>>                                                                    |   4%
  |
  |>>>                                                                   |   4%
  |
  |>>>                                                                   |   5%
  |
  |>>>                                                                   |   6%
  |
  |>>>>                                                                  |   6%
  |
```

```
  |>>>>                                                              |   7%
  |
  |>>>>>                                                             |   7%
  |
  |>>>>>                                                             |   8%
  |
  |>>>>>                                                             |   9%
  |
  |>>>>>>                                                            |   9%
  |
  |>>>>>>                                                            |  10%
  |
  |>>>>>>>                                                           |  10%
  |
  |>>>>>>>                                                           |  11%
  |
  |>>>>>>>                                                           |  12%
  |
  |>>>>>>>>                                                          |  12%
  |
  |>>>>>>>>                                                          |  13%
  |
  |>>>>>>>>                                                          |  14%
  |
  |>>>>>>>>>                                                         |  14%
  |
  |>>>>>>>>>                                                         |  15%
  |
  |>>>>>>>>>>                                                        |  15%
  |
  |>>>>>>>>>>                                                        |  16%
  |
  |>>>>>>>>>>                                                        |  17%
  |
  |>>>>>>>>>>>                                                       |  17%
  |
  |>>>>>>>>>>>                                                       |  18%
  |
  |>>>>>>>>>>>                                                       |  19%
  |
  |>>>>>>>>>>>>                                                      |  19%
  |
  |>>>>>>>>>>>>                                                      |  20%
  |
  |>>>>>>>>>>>>>                                                     |  20%
  |
  |>>>>>>>>>>>>>                                                     |  21%
  |
  |>>>>>>>>>>>>>                                                     |  22%
  |
```

```
  |>>>>>>>>>>>>>>>                                                   |  22%
  |
  |>>>>>>>>>>>>>>>                                                   |  23%
  |
  |>>>>>>>>>>>>>>>>                                                  |  23%
  |
  |>>>>>>>>>>>>>>>>                                                  |  24%
  |
  |>>>>>>>>>>>>>>>>                                                  |  25%
  |
  |>>>>>>>>>>>>>>>>>                                                 |  25%
  |
  |>>>>>>>>>>>>>>>>>                                                 |  26%
  |
  |>>>>>>>>>>>>>>>>>                                                 |  27%
  |
  |>>>>>>>>>>>>>>>>>>                                                |  27%
  |
  |>>>>>>>>>>>>>>>>>>                                                |  28%
  |
  |>>>>>>>>>>>>>>>>>>>                                               |  28%
  |
  |>>>>>>>>>>>>>>>>>>>                                               |  29%
  |
  |>>>>>>>>>>>>>>>>>>>                                               |  30%
  |
  |>>>>>>>>>>>>>>>>>>>>                                              |  30%
  |
  |>>>>>>>>>>>>>>>>>>>>                                              |  31%
  |
  |>>>>>>>>>>>>>>>>>>>>>                                             |  31%
  |
  |>>>>>>>>>>>>>>>>>>>>>                                             |  32%
  |
  |>>>>>>>>>>>>>>>>>>>>>                                             |  33%
  |
  |>>>>>>>>>>>>>>>>>>>>>>                                            |  33%
  |
  |>>>>>>>>>>>>>>>>>>>>>>                                            |  34%
  |
  |>>>>>>>>>>>>>>>>>>>>>>                                            |  35%
  |
  |>>>>>>>>>>>>>>>>>>>>>>>                                           |  35%
  |
  |>>>>>>>>>>>>>>>>>>>>>>>                                           |  36%
  |
  |>>>>>>>>>>>>>>>>>>>>>>>>                                          |  36%
  |
  |>>>>>>>>>>>>>>>>>>>>>>>>                                          |  37%
  |
```

```
  |>>>
```

```
|>>>
```

```
|>>>
```

```
|>>>
```

```
  |>>>
```

```
motivation. One unit increase in leadership is expected to be associated with
 an increase of 0.6198 units in autonomous motivation.

#Outcome Variable: control => path a2

#interpretation: About 59.42% of variance in controlled motivation is explain
ed by (transformational) leadership style.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Leadership significantly and negatively predicts controlled
motivation. One unit increase in leadership is expected to be associated with
 a decrease of 0.6025 units in controlled motivation.

#Outcome Variable: performance => path b1, b2, and c'

#interpretation: About 84.88% of variance in task performance is explained by
 leadership, autonomous motivation, AND controlled motivation

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation b1: Autonomous motivation significantly and positively predic
ts task performance. One unit increase in autonomous motivation is expected t
o be associated with an increase of 0.4890 units in task performance.

#interpretation b2: Controlled motivation significantly and negatively predic
ts task performance. One unit increase in controlled motivation is expected t
o be associated with an decrease of 0.2918 units in task performance.

#interpretation c': Leadership significantly and positively predict task perf
ormance. One unit increase in leadership is expected to be associated with an
 increase of 0.6782 units in task performance.

#Outcome Variable: performance => path c

#interpretation: About 79.80% of variance in task performance is explained by
 leadership ALONE.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation c: Leadership significantly and positively predicts task perf
ormance. One unit increase in leadership is expected to be associated with an
 increase of 1.1571 units in task performance.
```

```
#TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y
#Indirect effect(s) of X on Y => a1 x b1 AND a2 x b2
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a1 x b1: Autonomous motivation significantly mediates the rel
ation between leadership and task performance (a1b1 = 0.3031, 95% CI [.2443,
.3608]).

#interpretation a1 x b1: Leadership is significantly and indirectly related t
o task performance through the pathway of autonomous motivation (a1b1 = 0.303
1, 95% CI [.2443, .3608]).

#interpretation a2 x b2: Controlled motivation significantly mediates the rel
ation between leadership and task performance (a2b2 = 0.1758, 95% CI [.1199,
.2334]).

#interpretation a2 x b2: Leadership is significantly and indirectly related t
o task performance through the pathway of controlled motivation (a2b2 = 0.175
8, 95% CI [.1199, .2334]).

#Conclusion: When employees perceived their supervisors to be more transforma
tional, the higher autonomous motivation they have, which, in turn, is signif
icantly and positively associated with task performance. Also, when employees
 perceived their supervisors to be more transformational, the lower controlle
d motivation they have, which, in turn, is significantly and negatively assoc
iated with task performance.
```

Assignment 3

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**) serves as a causal antecedent variable, sugary drink consumption (**SugarDrink_21**) AND fruit consumption (**fruit_21**) serves as parallel mediators, and the condition of children's teeth (**TeethCond_21**) serves as a consequent variable. Please run a conditional process analysis to find out the answer. (Let's assume that these four variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 3_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.

- fruit consumption (**fruit_21**) : Higher values indicate that the child eat more fruits on a weekly basis.
- teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.

- Draw the diagram to represent the parallel multiple mediator model.
- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Run Pearson's bivariate correlations
- Run the conditional process analysis
- Find out the effect of path a1, a2, b1, b2, c, c', a1 x b1, and a2 x b2
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a1, a2, b1, b2, c, c', a1 x b1, and a2 x b2)
- Is your theory correct? Does sugary drink consumption significantly mediate the relation between age and teeth condition? Does fruit consumption significantly mediate the relation between age and teeth condition? Which path is more strongly associated with teeth condition?

Assignment 4

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**) serves as a causal antecedent variable, sugary drink consumption (**SugarDrink_21**), fruit consumption (**fruit_21**), and vegetable consumption (**vegetables_21**) serves as parallel mediators, and the condition of children's teeth (**TeethCond_21**) serves as a consequent variable. Please run a conditional process analysis to find out the answer. (Let's assume that these five variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 4_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.

- fruit consumption (**fruit_21**) : Higher values indicate that the child eat more fruits on a weekly basis.
- vegetable consumption (**vegetables_21**) : Higher values indicate that the child eat more vegetables on a weekly basis.
- teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.

- Draw the diagram to represent the parallel multiple mediator model.
- Write down your bmatrix
- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Run Pearson's bivariate correlations
- Run the conditional process analysis
- Find out the effect of path a1, a2, a3, b1, b2, b3, c, c', a1 x b1, a2 x b2, a3 x b3
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a1, a2, a3, b1, b2, b3, c, c', a1 x b1, a2 x b2, a3 x b3)
- Is your theory correct? Does sugary drink consumption significantly mediate the relation between age and teeth condition? Does fruit consumption significantly mediate the relation between age and teeth condition? Does vegetable consumption significantly mediate the relation between age and teeth condition? Which path is more strongly associated with teeth condition?

Assignment: Find an empirical journal article

- Find an empirical article in your field that uses the parallel multiple mediator model.
- What is the hypothesized model? That is, what is the diagram that visualizes the relations?
- Define X, M1, M2, and Y using layperson's terms
- Find out the values for a1, a2, b1, b2, c', a1 x b1, and a2 x b2.
- Find out R^2
- Share your interpretation of the results for each path (i.e., a1, a2, b1, b2, c', a1 x b1, a2 x b2) using layperson's terms

- Are the mediation effects significant? How do you know?
- Make a presentation to your classmates with PPT slides.
- Your presentation cannot go over than 5 mins.
- **Submit your PPT slides to the designated folder on iStudy.**